

8 June 2026
Jennifer Linnaea

CAHOOTS and the Welfare Check:

How Have Welfare Check Calls Changed Since CAHOOTS Stopped Service in Eugene?

DSCI 410 – Data Science in Action Research Project Report

Background

Welfare check calls are a category of call to Computer Aided Dispatch (CAD), often handled by law enforcement, that seeks to establish the wellbeing of a person. These calls can be phoned in by doctors, employers, neighbors, family members, friends, or concerned citizens.

CAHOOTS (Crisis Assistance Helping Out On The Streets), which provides mobile crisis response services in Springfield, Oregon and, until April 7th, 2025, provided these services in the neighboring city of Eugene, Oregon as well, historically handled about half of all welfare check calls in the two cities. The other half were handled by the Eugene and Springfield police departments in their respective cities.¹ Since Lane County's newer mobile crisis response service, Mobile Crisis Services of Lane County (MCSLC), does not respond to welfare check calls, my research was motivated by a desire to understand

¹ Eugene CAD Data 2015-2025, 2026 public record request by Dr. Rori Rohlf.
Springfield CAD Data 2015-2025, 2026 public record request by Dr. Rori Rohlf.

possible impacts of the CAHOOTS stoppage in Eugene on welfare check calls. Specifically, the research question this report seeks to answer is:

How have the number and dispatch rates of welfare check calls in Eugene changed since CAHOOTS ceased operation there?

Data

Data used for the analysis were CAD (Computer-Aided Dispatch) data from Eugene and Springfield, Oregon, spanning the years 2015 to 2025, and obtained by public records requests from Dr. Rori Rohlf, Associate Professor of Data Science at the University of Oregon. Each row in the data represented a single call for service, and the variables used were date, call nature, primary responding unit, and number/names of responding units.

Data Cleaning

The following data cleaning steps were taken:

Eugene CAD

- Yearly data imported and concatenated into a single data frame
- Dates converted to datetime objects
- Whitespace stripped
- Month and year columns created
- Unneeded columns dropped
- Rows with null values for primary unit dropped unless the number of units dispatched was zero

- Institution column created based on primary responding unit (CAHOOTS, EPD, neither)

Springfield CAD

The data cleaning process for Springfield CAD data was identical to the cleaning process for the Eugene data, with the following steps added:

- Primary unit data (received as a separate file) was left joined to the main Springfield data set using incident number as a key
- The number of responding units was calculated based on the units dispatched

Data strengths, limitations, and caveats

The strengths of the CAD data were consistent labelling and formatting over eleven years of call logs, and the low number of missing values in the variables I used. Primary limitations were the small time period of post-CAHOOTS data (about nine months), the possibility that dropped rows were not a random sample of calls, and the possibility that primary responding vehicle was not an accurate proxy for whether CAHOOTS or EPD/SPD answered a call.

All scripts for data cleaning are available on the author's GitHub:

<https://github.com/vaerynii/welfare-check-response-analysis>

Methods

Overview

To address the possibility that changes to welfare check call volume or dispatch rate in Eugene after CAHOOTS ceased operation there may have had unrelated causes, I used difference in differences (DiD) analysis, which compares the trends of two groups over time both before and after a “treatment” event: in this case, the end of CAHOOTS’ operations in Eugene. Because of the physical proximity and regional similarities of Eugene and Springfield, and because of the history of CAHOOTS’s work with law enforcement in both cities, Springfield (where CAHOOTS remains operational) was used as the control group.

In addition, because of population differences between the two cities,² all analyses were conducted using calls per 1000 people, adjusted by the population of each city in each year, rather than with raw call volumes.

Finally, to confirm that the pre-treatment parallel trends assumption necessary for difference in differences analyses was present, a parallel trends test was conducted before each difference in differences analysis. In each case, the test confirmed parallel trends pre-CAHOOTS cessation.

Resources

The packages used to run the analyses were:

² Eugene population was obtained from the City of Eugene website: <https://www.eugene-or.gov/5046/Population>

Springfield population was obtained from the World Population Review: <https://worldpopulationreview.com/us-cities/oregon/springfield>

- datetime
- numpy
- pandas
- matplotlib
- seaborn
- diff_diff

The inputs for each analysis were the cleaned Eugene and Springfield CAD data.

DiD Analysis 1: Total CAD Calls per 1000 People

Before testing whether welfare check call volumes changed in Eugene post-CAHOOTS, a preliminary difference in differences analysis was done to determine whether the **total** volume of population-adjusted CAD calls changed.

Data were prepared by grouping by month and year to create a new data frame with month, year, and call counts. To this new data frame, columns for population, calls per 1000 people, and period, as well as treatment group and pre/post treatment status were added. Two extreme outlier months that I suspected were the result of human error were removed from the data frame. Then a parallel trends test was run on the pre-April 2025 data.

After parallel trends test showed that parallel trends between the two cities existed beginning in 2019, a difference in differences analysis was run on 2019-2025 data. A line graph to visualize this data was also created.

DiD Analysis 2: Welfare Check Calls per 1000 People

The purpose of this analysis was to determine whether the number of population-adjusted welfare check calls changed in Eugene after CAHOOTS stopped service. For this analysis, the Eugene and Springfield CAD data sets were filtered to only include welfare check calls, but were otherwise prepared identically as in Analysis 1. Post-2019 parallel trends were confirmed, DiD analysis was run on the 2019-2025 data, and a line graph was created.

DiD Analysis 3: Police Dispatches to Welfare Check Calls per 1000 People

The third analysis was aimed at determining changes in Eugene police dispatch rates to welfare checks call after CAHOOTS. The data set of welfare check calls was filtered to only include EPD and SPD response, and then treated identically to Analysis 1. As with the previous two analyses, post-2019 parallel trends were confirmed, and a difference in differences analysis was run on the 2019-2025 population-adjusted police response to welfare checks data. A graph of these data was made.

DiD Analysis 4: Welfare Check Calls per 1000 People with No Dispatch

The purpose of the final analysis was to examine changes in no-dispatch rates for welfare check calls in Eugene after CAHOOTS stopped operating there. Data preparation was identical to Analysis 3, except that the data was filtered to only include welfare check calls that did not result in a unit being dispatched. In this case, parallel trends could only be established beginning in 2022. A DiD analysis was run on 2022-2025 no-dispatch welfare check data, and a line graph was created.

All analysis and graphing scripts are available on the author's GitHub:

<https://github.com/vaeryniil/welfare-check-response-analysis>

Results

Analysis 1 Results

Figure 1 shows the number of CAD calls per 1000 people in Eugene and Springfield from 2019 to 2025. A recent downward trend is visible in both cities, and the difference in differences analysis showed no significant difference in Eugene's call volume before and after CAHOOTS ceased operation (p-value=0.10).

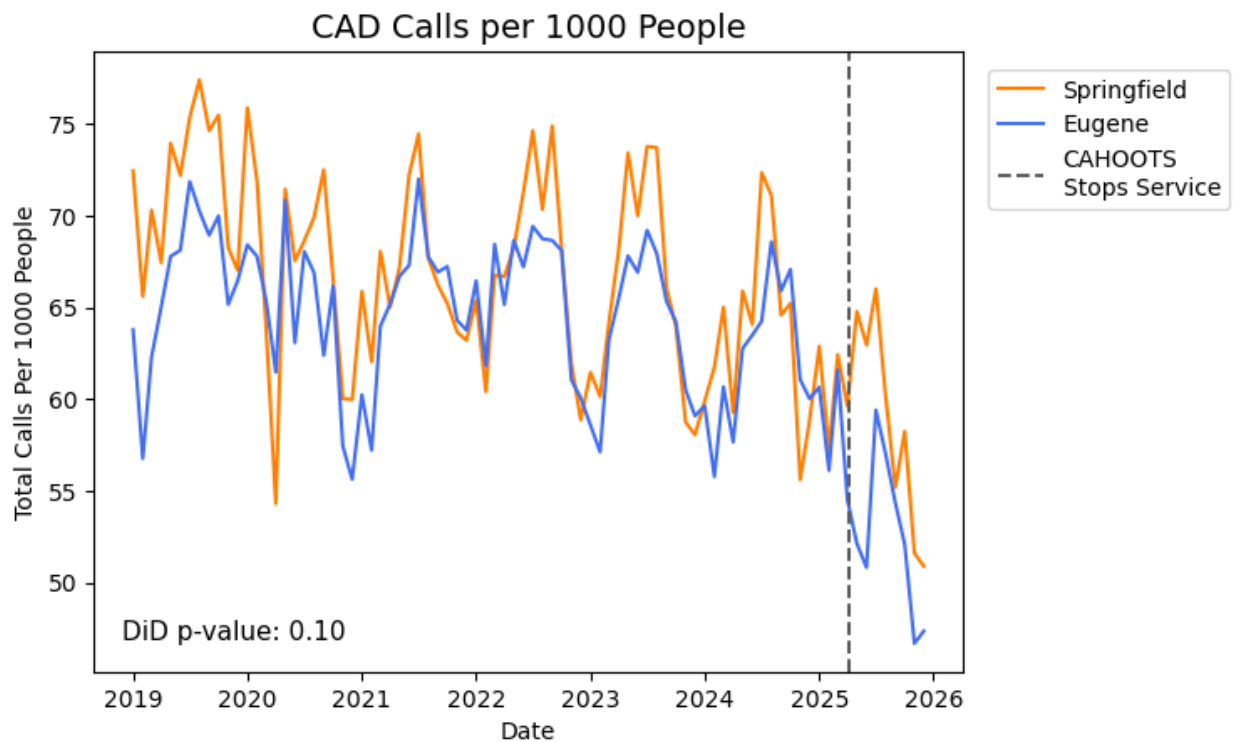


Figure 1: Total CAD Calls

Analysis 2 Results

In contrast, the DiD analysis of welfare check calls per 1000 people showed a significant difference ($p\text{-value}=0.0009$) between pre- and post-CAHOOTS welfare check call volume.

Figure 2 is a visualization of welfare checks per 1000 people over time for Eugene and Springfield. A decrease in welfare check calls in Eugene relative to Springfield after CAHOOTS ceased operation can be seen on the graph.

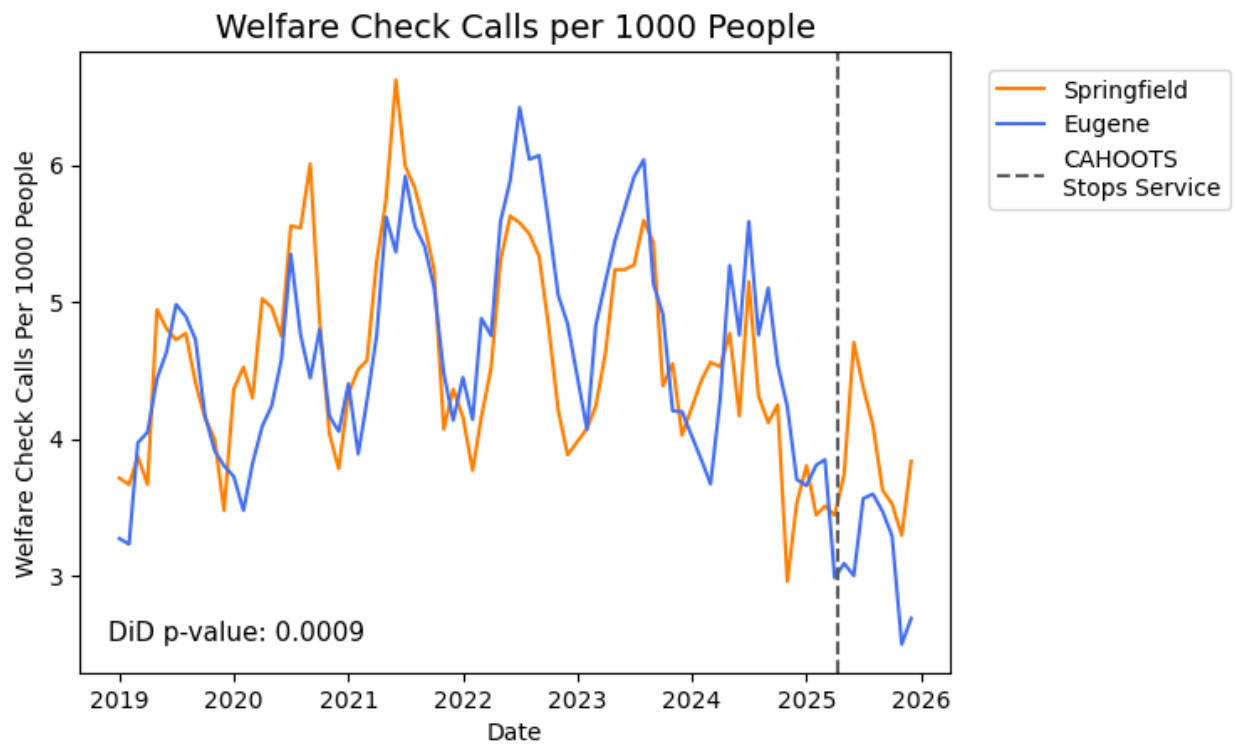


Figure 2: Welfare Check Calls

Analysis 3 Results

Figure 3 shows the population-adjusted number of welfare check calls answered by the Eugene and Springfield police departments from 2019 to 2025. While the number of calls responded to by the Eugene Police Department increased after CAHOOTS stopped service, the number of calls responded to by the Springfield Police Department increased as well. The difference in differences analysis showed no significant difference between the two groups (p-value=0.08).

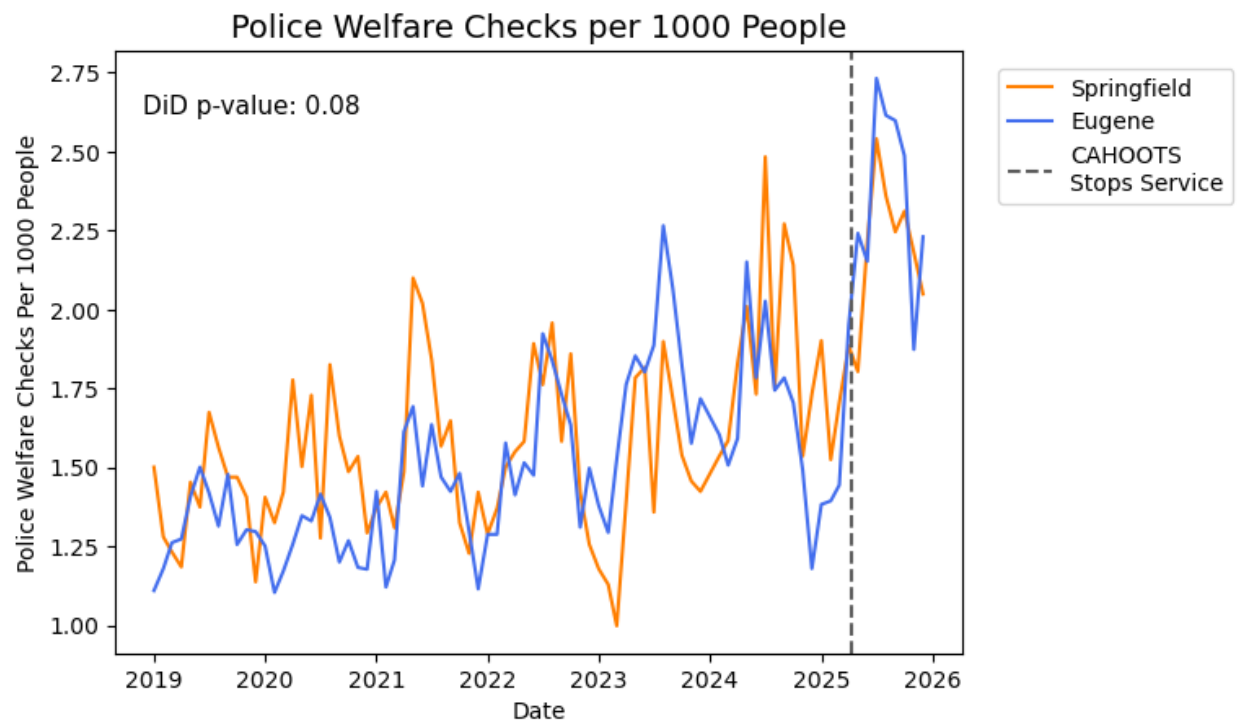


Figure 3: Police Response

Analysis 4 Results

Finally, the difference in differences analysis on population-adjusted welfare check calls for which no unit was dispatched showed a significant post-CAHOOTS difference in Eugene ($p\text{-value}=0.001$). Figure 4 shows the population-adjusted number of welfare check calls for which no unit was dispatched from 2022 to 2025. An increase in non-responses in Eugene relative to Springfield after CAHOOTS stopped operating is visible on this graph.

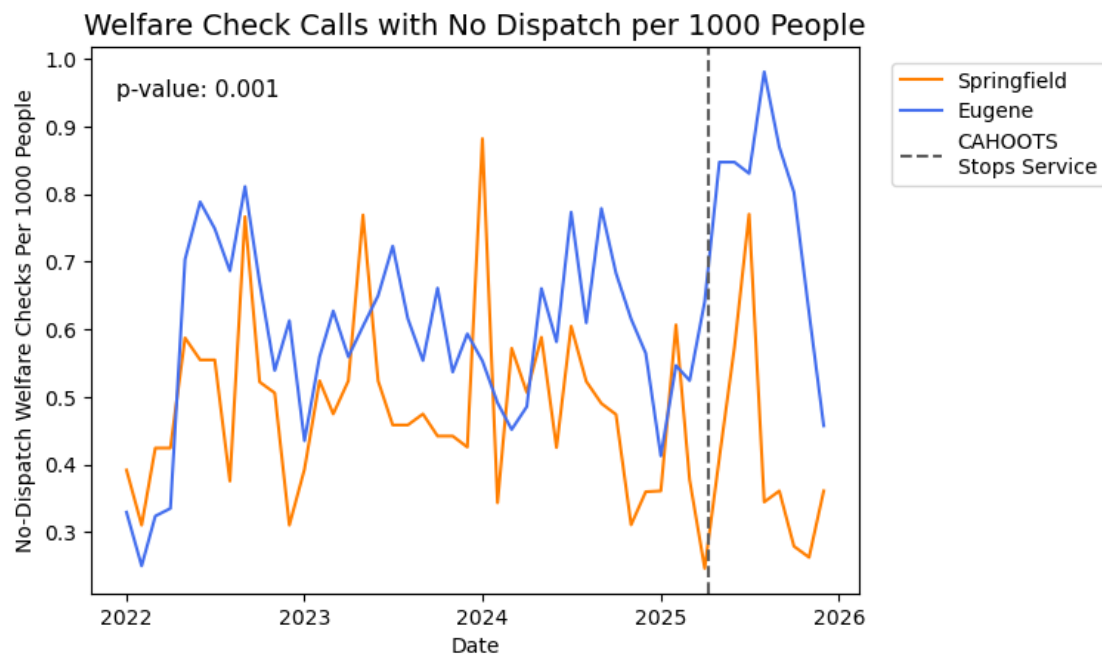


Figure 4: No Dispatch

Discussion

Findings

Through a series of differences in differences analyses, using the city of Springfield as a control, I sought to answer the question of how welfare check calls and their dispatch rates have changed in Eugene since CAHOOTS ceased operation. I found that while the total number of population-adjusted CAD calls were not significantly difference in Eugen post-CAHOOTS, the number of incoming welfare check calls was. In addition, while police response to welfare check calls did not go up significantly relative to Springfield, the number of unanswered welfare check calls did. This suggests two things: one, that fewer people are requesting welfare checks in Eugene since CAHOOTS stopped operating, and two, that more welfare check calls are going unanswered.

Strengths and Caveats

The pre-analysis parallel trends testing strengthens the results of this analysis, as does the population adjustment, without which Eugene's seasonal CAD trends appear more pronounced than Springfield's.

The main caveat to the results is the fact that only nine months of post-CAHOOTS data existed at the time of the public records request. Future research could take advantage of a growing body of post-CAHOOTS data to confirm or deny these findings.

Implications and Recommendations

If the findings are true, they represent a gap in service left by the defunding of CAHOOTS. If the City of Eugene determines that this is the case, they could begin closing this gap by first seeking to understand the reasons for the decrease in welfare check calls, as well as the reasons for decreased dispatch. While welfare check calls are a small percentage of CAD calls, I believe that they are important, as they are a way that we care for one other as a community.